EL HOUDAIGUI Othmane

Engineering Student (Apprentice) – Global Performance Management | AI & Predictive Maintenance

Arts et Métiers ParisTech — Grande École Program — Metz Campus

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LinkedIn Profile

Education

- **Arts et Métiers ParisTech** *2023 – 2026, Metz, France*
  Grande École Program – Apprenticeship in Global Performance Management
- **CPGE Lycée Mohammed V** *2021 – 2023, Casablanca, Morocco*
  Preparatory Classes for Top Engineering Schools (MPSI/MP)
- **Scientific Baccalaureate, Mathematics Specialization** *2021, Casablanca, Morocco*
  International Option, French Section Highest Honors

Experience

- **CAF — Construcciones y Auxiliar de Ferrocarriles**
  AI Engineering Apprentice — Railway Predictive Maintenance Sept. 2025 – Aug. 2026, Reichshoffen, France
  - Development of an **AI diagnostic assistant**.
  - Implementation of **unsupervised anomaly detection** (Isolation Forest, Autoencoders/VAE) on sensor data (real-time/status) using **windowing** and contextual aggregation (speed, altitude, conditions).
  - **Data Pipeline**: Multi-source ingestion, normalization, validation, and calculation of **health indices** per component.
- **RATP Group – Bus Maintenance Center** *July 2024 (1 month)*
  Internship in Mechanical Maintenance and Inventory Management Paris, France
  - Performed corrective and preventive maintenance tasks on engines and facilities.
  - Contributed to inventory control and stock database updates.

Academic Projects

- **AI Planner for Dynamic Production Optimization** *2024*
  Design of an adaptive scheduler integrating forecasting, optimization, and user interface
  - Development of an intelligent tool adjusting production schedules in real-time based on demand.
- **Mobile Base for an Autonomous UGV (Drone)** *2023*
  Project in partnership with SNCF for railway inspection
  - Mechanical design of the mobile base (CATIA V5).
  - Sizing and integration of electronic components.

Technical Skills

Management & Op. Excellence: KPI Design, Lean Six Sigma (PDCA, DMAIC, 5M, Kaizen, Kanban), Process Control
Programming: Python, R, MySQL, LaTeX
Data Analysis & AI: Neural Networks (ANN), ANOVA, Classification Techniques
Frameworks: scikit-learn, Flask
Tools: Power BI, Excel, Access, MATLAB, ABAQUS, 3DEXPERIENCE
Languages: Arabic (Native), French (C1 - Fluent), English (B2 - Professional Proficiency), Spanish (Beginner)

Certificates

- Coursera — *DeepLearning.AI : Deep Learning Specialization*
- FreeCodeCamp — *Scientific Computing with Python*

Interests

- Flight Simulation; Video Editing; Photography; Game Development with Unity
- Sports: Football, Swimming